

Campus Marketplace

Product Backlog — Software Design 2026
Prepared: 30 March 2026 | Backend: Firebase | Frontend: TBD

Epic	Sprint	Stories	Priority
E1: User Verification & Roles	Sprint 1	2	High
E2: Item Listings	Sprint 1	3	Medium
E3: Buyer Experience	Sprint 2	3	High
E4: In-App Messaging	Sprint 2	1	Medium
E5: Trade Facility Management	Sprint 2	3	Medium
E6: Payments	Sprint 2	3	Medium
E7: Rating & Trust System	Sprint 3	2	Medium
E8: SA Data Integration	Sprint 2	1	Medium
E9: Analytics & Reporting	Sprint 3	4	Low

E1 · User Verification & Roles [Sprint 1]

US1 — Student Registration & Login

High

As a student, I want to register and log in using a third-party identity provider so that my identity is verified and my account is secure.

Tasks

Task	Assignee	Status
Research & choose identity provider (Auth0 / Firebase Auth / Google OAuth)	TBD	To Do
Configure identity provider credentials and app settings	TBD	To Do
Build login and registration page UI	TBD	To Do
Integrate frontend auth flow with chosen provider	TBD	To Do
Store user profile in Firestore on first login	TBD	To Do
Implement route guards — redirect unauthenticated users to login	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a new student visits the site, **When** they click Sign Up and complete the identity provider flow, **Then** their account is created and they are redirected to the home page as a Student role.

Given a registered student is logged out, **When** they enter valid credentials, **Then** they are authenticated and land on their dashboard.

Given an unauthenticated user tries to access a protected page, **When** they navigate directly via URL, **Then** they are redirected to the login page.

US2 — Role Assignment

High

As an admin, I want to assign roles (Student, Trade Facility Staff, Admin) to users so that each user only has access to features appropriate to their role.

Tasks

Task	Assignee	Status
Define roles in Firestore user schema	TBD	To Do
Implement role-based access control (RBAC) using Firebase custom claims	TBD	To Do
Build admin UI to view users and assign or change roles	TBD	To Do
Write tests confirming Students cannot access Staff or Admin pages	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a logged-in admin opens the user management panel, **When** they assign a user the Trade Facility Staff role and save, **Then** that user gains staff feature access and loses student-only access.

Given a student user is authenticated, **When** they attempt to access the admin dashboard URL directly, **Then** they receive an Access Denied response.

E2 · Item Listings [Sprint 1]

US3 — Create an Item Listing

High

As a student seller, I want to create an item listing with photos, description, category, condition rating, and price so that other students can discover and purchase or trade my item.

Tasks

Task	Assignee	Status
Design listing creation form UI	TBD	To Do
Implement photo upload (min. 1, max. 5) using Firebase Storage	TBD	To Do
Build category dropdown (Textbooks, Electronics, Furniture, Clothing, Other)	TBD	To Do
Build condition rating selector (New, Like New, Good, Fair, Poor)	TBD	To Do
Add listing type toggle: For Sale / For Trade / Either	TBD	To Do
Save listing document to Firestore linked to seller UID	TBD	To Do
Show success confirmation and redirect to new listing page	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a logged-in student completes all required listing fields, **When** they submit the form, **Then** a new listing is created and visible on the marketplace.

Given a student creates a listing and uploads a photo, **When** the form is submitted, **Then** the photo is stored in Firebase Storage and displayed on the listing page.

Given a student submitting a listing leaves a required field empty, **When** they click Submit, **Then** the form shows a validation error and does not submit.

US4 — Edit or Remove a Listing

Medium

As a student seller, I want to edit or delete my own listings so that I can keep them accurate or remove items that are no longer available.

Tasks

Task	Assignee	Status
Show Edit and Delete buttons only to the listing owner	TBD	To Do
Build pre-populated edit form from Firestore data	TBD	To Do
Implement delete with confirmation dialog	TBD	To Do
Add Firestore security rules preventing non-owners from editing or deleting	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a logged-in seller views their own listing, **When** they click Edit, update the price, and save, **Then** the listing reflects the new price immediately.

Given a logged-in seller views their own listing, **When** they click Delete and confirm, **Then** the listing is removed from the marketplace.

Given a student who did not create a listing, **When** they attempt to access the edit URL directly, **Then** they are denied access with an error.

US5 — Mark Listing Type

Medium

As a student seller, I want to mark my listing as for sale, for trade, or either so that buyers know what kind of exchange I am open to.

Tasks

Task	Assignee	Status
Add listing type field to Firestore listing schema	TBD	To Do
Display listing type badge prominently on listing card and detail page	TBD	To Do
Allow filtering by listing type on the browse page	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a seller creates a listing and selects For Trade, **When** the listing is published, **Then** a For Trade badge is visible on the listing card and detail page.

Given a buyer browses listings and filters by For Sale, **When** the filter is applied, **Then** only listings marked For Sale or Either are shown.

E3 · Buyer Experience [Sprint 2]

US6 — Search and Filter Listings

High

As a student buyer, I want to search and filter listings by category, condition, and price so that I can quickly find items I am looking for.

Tasks

Task	Assignee	Status
Build search bar with Firestore full-text or Algolia integration	TBD	To Do
Implement category, condition, and price range filter controls	TBD	To Do
Display filtered results in responsive listing grid	TBD	To Do
Show empty state when no results match	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a buyer is on the browse page, **When** they type Calculus into the search bar, **Then** only listings with Calculus in the title or description are shown.

Given a buyer filters by Electronics and condition New, **When** the filters are applied, **Then** only New Electronics listings are returned.

Given no listings match the search query, **When** the buyer submits the search, **Then** a friendly no results message is displayed.

US7 — View Seller Rating

Medium

As a student buyer, I want to view a seller's rating and transaction history so that I can assess their trustworthiness before buying.

Tasks

Task	Assignee	Status
Display average star rating and review count on seller profile	TBD	To Do
Show completed transaction count on seller profile	TBD	To Do
Link seller name on listing page to their profile	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a buyer views a listing, **When** they click on the seller's name, **Then** the seller's profile opens showing their rating and completed transaction count.

US8 — Initiate Purchase or Trade Offer

High

As a student buyer, I want to initiate a purchase or trade offer on a listing so that I can begin the transaction process.

Tasks

Task	Assignee	Status
Add Buy Now and Make Trade Offer buttons on listing detail page	TBD	To Do
Create Firestore transaction document on offer initiation	TBD	To Do
Notify seller of new offer via in-app notification	TBD	To Do

Task	Assignee	Status
Allow seller to accept or decline the offer	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a logged-in buyer views a For Sale listing, **When** they click Buy Now, **Then** a transaction is created and the seller receives a notification.

Given a seller receives an offer, **When** they click Accept, **Then** the transaction status updates to Accepted and the buyer is notified.

E4 · In-App Messaging [Sprint 2]

US9 — Buyer-Seller Chat

Medium

As a student, I want to chat with a buyer or seller through the platform so that I can negotiate and arrange transaction details safely without sharing personal contact information.

Tasks

Task	Assignee	Status
Design chat UI (message list and input box)	TBD	To Do
Implement real-time chat using Firestore or Firebase Realtime Database	TBD	To Do
Restrict chat to the two parties involved in a transaction	TBD	To Do
Add unread message badge indicator	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a buyer initiates an offer, **When** the seller opens the transaction chat, **Then** both parties can send and receive messages in real time.

Given a student tries to access a chat they are not part of, **When** they attempt to open the chat URL, **Then** they are denied access.

E5 · Trade Facility Management [Sprint 2]

US10 — Book Drop-Off Slot

Medium

As a student, I want to book a drop-off time slot at the trade facility so that I can securely hand over an item I have sold.

Tasks

Task	Assignee	Status
Display available time slots for the trade facility	TBD	To Do
Allow students involved in an accepted transaction to book a slot	TBD	To Do
Store booking in Firestore and prevent double-booking	TBD	To Do
Send booking confirmation notification to student	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a student has an accepted transaction, **When** they select an available slot and confirm, **Then** the slot is booked and a confirmation notification is sent.

Given a slot is already fully booked, **When** a student tries to book it, **Then** the slot shows as unavailable and cannot be selected.

US11 — Staff Confirm Item Receipt and Release

Medium

As trade facility staff, I want to confirm receipt and release of items so that transactions are properly tracked and both parties are protected.

Tasks

Task	Assignee	Status
Build staff dashboard showing upcoming drop-offs and collections	TBD	To Do
Add Confirm Drop-Off and Confirm Collection buttons per transaction	TBD	To Do
Update Firestore transaction status on each confirmation	TBD	To Do
Notify both parties when item status changes	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given staff receives an item at the facility, **When** they click Confirm Drop-Off, **Then** the transaction status updates and both students are notified.

Given staff releases an item to the buyer, **When** they click Confirm Collection, **Then** the transaction is marked complete.

US12 — Configure Facility Hours and Capacity

Low

As an admin, I want to configure trade facility operating hours and slot capacity so that the facility runs efficiently and students can only book valid times.

Tasks

Task	Assignee	Status
Build admin settings UI for operating hours and slots per hour	TBD	To Do
Store facility config in Firestore and apply to slot availability logic	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given an admin updates the facility to close at 16:00, **When** a student tries to book a 17:00 slot, **Then** the slot is not available.

E6 · Payments [Sprint 2]

US13 — Online Payment

High

As a student buyer, I want to pay for an item online via a third-party payment gateway so that the transaction is completed securely.

Tasks

Task	Assignee	Status
Research and select payment gateway (PayFast / Yoco / Stripe)	TBD	To Do
Integrate payment gateway checkout into transaction flow	TBD	To Do
Record payment status in Firestore on webhook confirmation	TBD	To Do
Display payment confirmation to buyer and seller	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a buyer accepts a transaction and proceeds to payment, **When** they complete the payment gateway flow, **Then** the transaction is marked Paid and both parties are notified.

Given a payment fails, **When** the gateway returns an error, **Then** the transaction remains unpaid and the buyer sees an error message.

US14 — Partial Payment with Cash Shortfall

Medium

As a student buyer, I want to pay a partial amount online and have the cash shortfall recorded so that I can settle the remainder in person at the trade facility.

Tasks

Task	Assignee	Status
Allow buyer to enter a partial payment amount at checkout	TBD	To Do
Record online amount paid and outstanding cash shortfall in Firestore	TBD	To Do
Display outstanding cash amount clearly to trade facility staff	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a buyer pays R300 of a R500 item online, **When** the payment is confirmed, **Then** the transaction shows R200 outstanding cash shortfall.

US15 — Staff Confirm Cash Shortfall Settled

Medium

As trade facility staff, I want to confirm that any cash shortfall has been paid before releasing an item so that sellers always receive their full payment.

Tasks

Task	Assignee	Status
Show outstanding cash amount on staff transaction view	TBD	To Do
Add Confirm Cash Received button, locked until item is present	TBD	To Do
Update Firestore transaction to Fully Paid on confirmation	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a transaction has an outstanding cash shortfall, **When** staff confirm cash received and click release, **Then** the transaction is marked Fully Paid and the item can be released.

E7 · Rating & Trust System [Sprint 3]

US16 — Leave Rating and Review

Medium

As a student, I want to leave a star rating and written review after a completed transaction so that others can make informed decisions about trading with that person.

Tasks

Task	Assignee	Status
Prompt both parties to rate after transaction is marked complete	TBD	To Do
Build star rating and text review submission form	TBD	To Do
Store review in Firestore and recalculate average rating for user	TBD	To Do
Display reviews on user profile page	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a transaction is marked complete, **When** both parties are prompted to leave a review, **Then** each can submit a star rating and written review.

Given a user receives a new review, **When** the review is submitted, **Then** their average rating is updated immediately on their profile.

US17 — Moderate Abusive Content

Low

As an admin, I want to flag or remove abusive reviews so that the platform remains safe and trustworthy for all users.

Tasks

Task	Assignee	Status
Add Report button on each review visible to all users	TBD	To Do
Build admin moderation queue showing reported reviews	TBD	To Do
Allow admin to remove review or dismiss report	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a user reports a review as abusive, **When** an admin views the moderation queue, **Then** the reported review appears and can be removed or dismissed.

E8 · SA Data Integration [Sprint 2]

US18 — SA Price Suggestions for Listings

Medium

As a student seller, I want to see a suggested price for my listing based on publicly available South African consumer price data so that I can price my item competitively and fairly.

Tasks

Task	Assignee	Status
Research suitable SA public dataset (Stats SA CPI, SARB, or other)	TBD	To Do
Document chosen data source and justify its reliability in the backlog	TBD	To Do
Build data ingestion script or API call to fetch current price index	TBD	To Do
Map CPI categories to listing categories and compute price suggestion	TBD	To Do
Display suggested price range on listing creation form	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given a seller creates an Electronics listing, **When** the form loads, **Then** a suggested price range based on SA consumer price data is displayed.

Given the SA data source is updated, **When** the ingestion job runs, **Then** the displayed price suggestions reflect the updated data.

E9 · Analytics & Reporting [Sprint 3]

US19 — Popular Categories Report

Low

As an admin, I want to view a report of the most popular item categories and completed transactions over time so that I can understand platform usage trends.

Tasks

Task	Assignee	Status
Aggregate completed transaction data from Firestore by category	TBD	To Do
Build bar or line chart dashboard component	TBD	To Do
Add date range filter to the report	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given an admin opens the analytics dashboard, **When** they select a date range, **Then** a chart showing completed transactions per category for that period is displayed.

US20 — Trade Facility Utilisation Report

Low

As an admin, I want to view a trade facility utilisation report so that I can manage slot capacity effectively.

Tasks

Task	Assignee	Status
Aggregate slot bookings vs capacity from Firestore	TBD	To Do
Display utilisation percentage per day and time slot	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given an admin views the facility report, **When** all bookings for the week are loaded, **Then** utilisation percentage is shown per day and time slot.

US21 — Moderated Content Summary Report

Low

As an admin, I want to view a summary of flagged and moderated content so that I can monitor platform safety over time.

Tasks

Task	Assignee	Status
Count reported and removed reviews by period	TBD	To Do
Display summary table on admin dashboard	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given an admin opens the moderation report, **When** the data loads, **Then** a table showing total reports, removals, and dismissals for the selected period is shown.

US22 — Export Reports as CSV or PDF

Low

As an admin, I want to export any dashboard report as a CSV or PDF so that I can share data with stakeholders offline.

Tasks

Task	Assignee	Status
Add Export as CSV button to each report using client-side CSV generation	TBD	To Do
Add Export as PDF button using a print stylesheet or PDF library	TBD	To Do

User Acceptance Tests (Given-When-Then)

Given an admin views a report, **When** they click Export as CSV, **Then** a CSV file is downloaded containing the report data.
Given an admin views a report, **When** they click Export as PDF, **Then** a formatted PDF of the report is downloaded.